

Manual

SMA Posting Functions in Material Management SMA-MMG 8.2

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1 Material management

General

App name	Material management
Short name of app	Material management
Function authorization	sma.materialman

The application *material management* provides the following functions:

- Batch information
- Repost Batch
- Generate batch
- Stock overview
- Pool batches
- Split batch
- Assign batch
- [Unpack handling unit \(selected\)](#)
- [Unpack handling unit \(all\)](#)
- [Pack handling unit](#)

Function calls

The application provides the following function calls:

Batch information

The *batch information* shows relevant information on a batch (e.g. material number, batch class, quantity).

Repost Batch

Use this function to repost a batch from one material buffer or storage location to another.

Generate batch

Use this function to generate a new batch and store information about the batch. HYDRA automatically generates the batch number.

Stock overview

The *stock overview* provides an overview of the current storage locations. It specifies which batches are currently stored in a selected material buffer or storage location. The stock overview also informs about the current batch status.

Pool batches

Use the function call *Merge batches* to merge several batches into a new batch. The system automatically assigns a new and unique batch number to the new batch.

Split batch

Use the function call *Split batch* to split a batch into several individual batches. Depending on the selected mode,

- the original batch is reduced by the split quantity or
- the remaining quantity of the original batch is posted to a new batch.

Assign batch

Use the function call *Assign batch* to add a batch to a transport unit at a packing station.

Unpack handling unit (selected)

Use the function call *Unpack handling unit (selected)* to unpack a batch that is included in a handling unit.

Unpack handling unit (all)

Use the function call *Unpack handling unit (all)* to unpack all batches that are included in a handling unit.

Pack handling unit

Use the function call *Pack handling unit* to pack a batch into a selected handling unit.

2 Batch information

General

App name	Batch information
Short name of app	Batch info
Function authorization	sma.batchinfo

You can use the application *batch information* to view information about a batch used in HYDRA. You can enter the batch number manually or by scanning.

Selection criteria

The application provides the following selection criteria:

Batch number

Enter the number of the batch whose information you want to view.

You can enter the batch number manually or by scanning.

Fields

Batch number

Shows the batch number entered manually or by scanning.

Material number

Shows the material number of the entered batch.

Description

Shows the batch name.

Batch class

Shows the batch class. By default, the following batch classes are possible:

- Yield
- Scrap
- Rework
- Open quantity

Quantity

Shows the current quantity of the entered batch.

Batch status

Shows the current batch status of the entered batch. By default, the following batch statuses are possible:

- Free
- Blocked
- Running
- Processed

Manufacturing date

Shows the manufacturing date of the entered batch. The manufacturing date indicates the point in time when the batch was generated.

Material buffer

Shows the material buffer of the entered batch or the buffer currently including the batch.

Machine

Shows the machine/workplace where the batch is logged on or where the batch was generated.

OP

Shows the operation to which the batch is logged on or where the batch was generated.

3 Generate batch

App name	Generate batch
Short name of app	Generate batch
Function authorization	sma.createbatch

General

You can generate a batch by starting the function *Generate batch*. The system automatically assigns a distinct batch number. You can use the function for incoming goods, etc.

Fields

The function provides the following fields:

Material number

Shows the material number of the batch to be generated.

Material type

Shows the material type of the batch to be generated. You can enter the material type or choose it from the selection menu.

Quantity

Shows the quantity of the batch to be generated.

Unit

Shows the quantity unit of the batch to be generated.

Material buffer

Shows the material buffer that should include the generated batch.

Batch class

Shows the batch class of the batch to be generated. You can assign the following batch classes:

- Yield
- Scrap
- Rework
- Open quantity

Reason

Shows the reason configured in HYDRA for each batch class. You can enter the appropriate scrap reason for the batch class *scrap*. You can enter the reason manually or choose it from the selection menu.

Transport unit

Shows the transport unit of the batch to be generated. You can enter the transport unit manually or choose it from the selection menu.

Staff badge number

You have to enter your staff badge number in order to execute the function.

Buttons

Generate batch



Do not generate batch

Result

The system issues the following message, once the batch has been generated successfully:
"Ok! Batch xxx has been generated."

4 Repost batch

App name	Repost batch
Short name of app	Repost batch
Function authorization	sma.repostbatch

General

The function *repost batch* allows you to repost a batch from one material buffer or storage location to another.

Fields

The function provides the following fields:

Batch number

Enter the number of the batch whose information you want to view.

You can enter the batch number manually or by scanning.

Target buffer

Material buffer where the batch is transferred to. You can enter the target buffer manually or via the search screen.

Comment

You can add a brief comment about reposting and store it with the batch.

Staff badge number

You have to enter your staff badge number in order to execute the function.

Buttons



Repost batch



Do not repost batch

Result

The system issues the following message, once the batch has been reposted successfully:
"Ok! Batch xxx has been reposted."

5 Stock overview

General

App name	Stock overview
Short name of app	Stock overview
Function authorization	sma.stockview

You can use the application *stock overview* to gain insight into the material available in material buffers. Different statuses classify the material available in material buffers. You are informed about:

- The amount of material you can use
- The amount of material that still needs to be checked in the buffer.
- The amount of material in a buffer. The amount of blocked and unusable material.

Selection criteria

Material buffer

The size of the used device specifies how material buffers are aligned (responsive design). Small devices (e.g. smartphones) show material buffers as tabs above the evaluation list. Otherwise, material buffers are listed at the left margin of the application. If you select the required material buffer, all pieces of information (e.g. material, quantity, status) are shown.

Fields

Material number

The field *material* includes the material number of the material available in the material buffer.

Quantity

The field *quantity* shows the total amount of material included in the material buffer sorted by material number and batch status.

Batch status

The stock overview uses the following status categories and icons to indicate available material:



processed, deleted, completed, blocked, expired



delivered, running, minimum storage period, returned, advised material, transport



Inspection



Free

You can summarize batches based on their material number and status category. The current batch status is shown, once you've clicked the status category (icon).

6 Assign batch

Overview

App name	Assign batch
Short name of app	Assign batch
Function authorization	sma.ceanpa

You can use the function call *Assign batch* to add a batch to a transport unit at a packing station.

Requirements

The requirements for the use of packing functionalities are included in the document [Activating Palletizing Packaging.pdf](#).

Fields

The function *Assign batch* provides the following fields:

Workplace

In the field *Workplace*, enter the number of the workplace where you pack the handling unit.

Operation

Once you have selected a workplace, you can use the search help to display the list of logged on operations. Use the search help to select an operation.

Transport unit

Once you have selected an operation, the transport unit currently logged on to this operation is transferred into the field *Transport unit*.

Batch number

In the field *Batch number*, enter the batch number that you want to pack into the transport unit.

Number of batches

The field *Number of batches* shows the number of batches that are currently packed into the transport unit.

Total quantity

The field *Total quantity* shows the total quantity of batches in the transport unit.

Staff badge number

To add or to remove a batch to or from a transport unit, you must enter a valid staff badge number.

Tables

Lots

If you have selected a workplace and an operation, and if batches exist in the currently logged on transport unit, the following batch data is displayed:

- Batch number
- Article
- Quantity
- Time
- Article name

Buttons

Add

If a batch is entered in the field *Batch number*, the system packs the batch into the transport unit if you click the button *Add*.

Remove

If you have selected a batch in the table and entered a valid staff badge number, the system removes the selected batch from the current transport unit.

Complete transport unit

Click the button *Complete transport unit* to open the function call *Complete transport unit*. Use the function call *Complete transport unit* to complete the transport unit.

7 Complete transport unit

Overview

App name	Complete transport unit
Short name of app	Complete transport unit
Function authorization	sma.cawlpa

You can use the function call *Complete transport unit* to complete a transport unit at a packing station.



You cannot call the function *Complete transport unit* from the category material management. You call the function *Complete transport unit* using the function call [Assign batch](#).

Fields

The function call *Complete transport unit* provides the following fields:

Workplace

When you call the function *Complete transport unit*, the system takes over the respective workplace number. This number is entered in the field *Workplace*.

MES order number

Use the search help to select an order. You then complete the transport unit for this order.

Article

If you have selected an MES order number, the field *Article* shows the article of the operation.

New HU

If you have selected an MES order number, the field *New HU* shows the new number of the transport unit (handling unit).

Currently logged in batch

If you have selected an MES order number, the field *Currently logged in batch* shows the number of the currently logged on transport unit.

Target buffer

In the field *Target buffer*, enter the material buffer where the transport unit will be stored when you complete the function call.

Batch class

In the field *Batch class*, you specify the batch class that the transport unit will have when you complete the function call.

Transport unit

In the field *Transport unit*, you specify the type of transport unit.

Number of batches

If you have selected an MES order number, the field *Number of batches* shows the number of batches that are currently included in the transport unit.

Status

In the field *Status*, you specify the status that the transport unit will have when you complete the function call.

Tare weight (KG)

In the field *Tare weight (KG)*, you enter the tare weight of the transport unit.

Net weight (KG)

In the field *Net weight (KG)*, you enter the net weight of the transport unit.

Gross weight

The system calculates the value in the field *Gross weight* using the values entered in the fields *Tare weight (KG)* and *Net weight (KG)*. The sum total of the values specified is entered in the field *Gross weight*.

Reason

If the batch class is not G (yield), you can enter a reason in the field *Reason* and specify why the batch class is different.

Buttons**Cancel**

The function call is closed without any further action.

Generate HU

The transport unit (handling unit) is completed and a new transport unit is generated for the selected MES order number.

8 Pack handling unit

Overview

App name	Pack handling unit
Short name of app	Pack handling unit
Function authorization	sma.ceanpack

You can use the function call *Pack handling unit* to pack a batch into a selected handling unit.

Fields

The function call *Pack handling unit* provides the following fields:

Workplace

In the field *Workplace*, enter the number of the workplace where you pack the handling unit.

Handling unit

In the field *Handling unit*, enter the number of the handling unit that you want to use to include a batch. The number of the handling unit is not assigned automatically.

Batch

In the field *Batch*, enter the batch number of the batch that you want to pack into the handling unit.

Tables

Header data

If you have entered the number of a handling unit in the field *Handling unit*, the table *Header data* shows the following handling unit data:

- Article
- Total quantity
- Person
- Order
- Packing date
- Packing time

Detail data

If you have entered the number of a handling unit in the field *Handling unit*, the table *Detail data* shows the following data for batches that are included in the handling unit:

- Batch
- Internal batch number
- Article
- Quantity
- Current status

Buttons

Cancel

The function call is closed without any further action.

Pack handling unit

The batch specified in the field *Batch* is packed into the handling unit specified in the field *Handling unit*.

9 Unpack handling unit (all)

Overview

App name	Unpack handling unit (all)
Short name of app	Unpack handling unit (all)
Function authorization	sma.cedelpa

You can use the function call *Unpack handling unit (all)* to unpack all batches that are included in a handling unit.

Fields

The function call *Unpack handling unit (all)* provides the following fields:

Handling unit

In the field *Handling unit*, enter the number of the handling unit that you want to unpack. All batches included in the specified handling unit are unpacked.

Tables

Header data

If you have entered the number of a handling unit in the field *Handling unit*, the table *Header data* shows the following data for the handling unit:

- Article
- Total quantity
- Person
- Order
- Packing date
- Packing time

Detail data

If you have entered the number of a handling unit in the field *Handling unit*, the table *Detail data* shows the following data for batches that are included in the handling unit:

- Batch
- Internal batch number
- Article
- Quantity

- Current status

Buttons

Cancel

The function call is closed without any further action.

Unpack all

All batches included in the handling unit are unpacked.

10 Unpack handling unit (selected)

Overview

App name	Unpack handling unit (selected)
Short name of app	Unpack handling unit (selected)
Function authorization	sma.cedelpa

You can use the function call *Unpack handling unit (selected)* to unpack a batch that is included in a handling unit.

Fields

The function call *Unpack handling unit (selected)* provides the following fields:

Handling unit

In the field *Handling unit*, enter the number of the handling unit that includes the batch that you want to unpack.

Tables

Header data

If you have entered the number of a handling unit in the field *Handling unit*, the table *Header data* shows the following data for the handling unit:

- Article
- Total quantity
- Person
- Order
- Packing date
- Packing time

Detail data

If you have entered the number of a handling unit in the field *Handling unit*, the table *Detail data* shows the following data for batches that are included in the handling unit:

- Batch
- Internal batch number
- Article

- Quantity
- Current status

In the table *Header data*, select the batch number of the batch that you want to unpack .

Buttons

Cancel

The function call is closed without any further action.

Unpack all

The selected batch is unpacked from the handling unit.

11 MPL and TRT collection functions

Overview

This document describes the MPL and TRT posting functions of the SMA application *Data collection*.

If you use the MPL and TRT functions in the function group *Order*, additional input fields are available in the following posting functions:

- [Log operation on](#)
- [Log operation off](#)
- [Interrupt operation](#)

The function group *Material* provides the following posting functions:

- [Log input batch on](#)
- [Log input batch off](#)
- [Log on input batch in advance](#)
- [Log off reserved batch](#)
- [Change output batch](#)

Log operation on

Function authorization	sma.logon
------------------------	-----------

Use the dialog *Log operation on* to log an operation on to the workplace selected in the list on the left.

If you call the function, an intermediate dialog opens. The dialog displays the operations planned for the workplace. To display the operations, the application refers to the option *Sequencing list* in the *Workplace configuration*. Select the required operation to open the input dialog.



If the operation requires batch management, the following fields are displayed.

Input fields

Workplace

The system enters the selected workplace.

MES order number

The system enters the MES order number of the operation that you have selected in the intermediate dialog that opened on calling the function.

Components

In the field *Components*, enter the BOM item of the selected component.

Input batch

In the field *Input batch*, enter the batch number of the input batch you want to log on.

Status

Enter the workplace/machine status manually or select a status from the list that you can open using the magnifier symbol.

Buttons

Cancel

The button *Cancel* closes the dialogue without any further action.

Log on

The button *Log on* logs the operation on to the workplace. During operation logon, the standard validation checks of HYDRA are performed.

If the processing of the staff badge number is enabled, the validation check also includes personal data.



The person who logs on the operation is *not* automatically logged on with the operation. You can configure that the person who logs on the operation is logged on with the operation. To do so, assign the function authorization *sma.logonwpers* to the person.



In case of operations that require batch management, you can only log on one component during OP logon. If other components exist, you must logon these components using the function *Log input batch on*.

Log operation off

Function authorization	sma.logoff
------------------------	------------

Use the dialog *Log operation off* to log a logged on operation off from the currently selected workplace.



If the operation requires batch management, the following fields are displayed.

Input fields

Workplace

The system enters the selected workplace.

MES order number

The system enters the MES order number of the operation logged on to the workplace. If several operations are logged on to the workplace, you can select a different MES order number using a search help.

Material

The material number of the currently logged on output batch is displayed.

Currently logged on batch

The batch number of the currently logged on output batch is displayed.

Target buffer

This is the material buffer for the posting of the produced output batch. The search help provides a list of the material buffers.

Transport unit

Transport unit of the produced output batch. The search help provides a list of transport units.

Info on batch

You can enter a comment on the currently logged on output batch in the input field *Info on batch*.

Batch class

Here, the user can display a list of possible batch statuses for the output batch. The user can select the batch status from the list.

Yield

Input field for the produced yield in the primary quantity unit of the operation.

Scrap

Input field for the produced scrap in the primary quantity unit of the operation.



The system can only support the input of yield OR scrap. If you enter values in both fields, this can lead to processing problems.

Reason

Enter a scrap reason. The predefined value in this field is 0.

You can only enter a scrap reason that has previously been configured. The search function provides a selection. If you empty the field, the scrap is recorded without scrap reason.

Status

Here, the user can display a list of available machine statuses. The user can select a new status from the list.

Buttons**Cancel**

The button *Cancel* closes the dialogue without any further action.

Log off

The button *Log off* logs the operation off. During logoff, the standard validation checks might be performed (e.g. check for over/underdelivery).

If processing of the staff badge number is enabled, the validation check also includes personal data.

Interrupt operation

Function authorization	sma.interrupt
------------------------	---------------

Use the dialog *Interrupt operation* to interrupt a logged on operation at the currently selected workplace.



If the operation requires batch management, the following fields are displayed.

Workplace

The system enters the selected workplace.

MES order number

The system enters the MES order number of the operation logged on to the workplace. If several operations are logged on to the workplace, you can select a different MES order number using a search help.

Material

The material number of the currently logged on output batch is displayed.

Currently logged on batch

The batch number of the currently logged on output batch is displayed.

Target buffer

This is the material buffer for the posting of the produced output batch. The search help provides a list of the material buffers.

Transport unit

Transport unit of the produced output batch. The search help provides a list of transport units.

Info on batch

You can enter a comment on the currently logged on output batch in the input field *Info on batch*.

Batch class

You can display a list of possible batch statuses for the output batch. Select the batch status from the list.

Yield

Input field for the produced yield in the primary quantity unit of the operation.

Scrap

Input field for the produced scrap in the primary quantity unit of the operation.



The system can only support the input of yield OR scrap. If you enter values in both fields, this can lead to processing problems.

Reason

Enter a scrap reason. The predefined value in this field is 0.

You can only enter a scrap reason that has previously been configured. The search function provides a selection. If you empty the field, the scrap is recorded without scrap reason.

Status

You can display a list of possible statuses for the machine. Select the new status from the list.

Buttons**Cancel**

The button *Cancel* closes the dialogue without any further action.

Interrupt

This button interrupts the operation. The standard validation checks might be performed (e.g. check for overdelivery).

If processing of the staff badge number is enabled, the validation check also includes personal data.

Log input batch on

Function authorization	sma.cean
------------------------	----------

You use the posting dialog *Log input batch on* to log on an input batch to a BOM item of the current operation.

If you call the function, an intermediate dialog opens. This intermediate dialog shows the component list of the operation. Enter a batch number to log the input batch on to a selected BOM item.

Input fields

MES order number

The system enters the MES order number of the operation logged on to the workplace. If several operations are logged on to the workplace, you can select a different MES order number using a search help.

Input batch

In the field *Input batch*, enter the batch number of the input batch you want to log on.

Buttons

Cancel

The button *Cancel* closes the dialogue without any further action.

Log on

The input batch is logged on to a BOM item of the component list.

Log input batch off

Function authorization	sma.ceab
------------------------	----------

You use the posting dialog *Log input batch off* to log off an input batch from a current operation.

If you call the function, an intermediate dialog opens. This intermediate dialog shows the component list of the operation. Select an entry in the component list to specify the input batch that you want to log off.

Input fields

Input batch

In the field *Input batch*, enter the batch number of the input batch you want to log off.

Batch status

In the field *Batch status*, specify the batch status that the input batch will have after logoff. You can directly enter the batch status or select a status using the search help.

Consumption

In the field *Consumption*, enter the consumption of the input batch that you want to log off.

Buttons

Cancel

The button *Cancel* closes the dialogue without any further action.

Log off

The selected input batch is logged off

Log on input batch in advance

Function authorization	sma.cevan
------------------------	-----------

You use the posting dialog *Log on input batch in advance* to log on an input batch for a current operation in advance. Using this advance logon, the input batch logged on in advance is automatically logged on to the respective BOM item if the input batch changes.

If you call the function, an intermediate dialog opens. This intermediate dialog shows the component list of the current operation. Select an entry in the component list to log on the input batch in advance to this BOM item.

Input fields

MES order number

The system enters the MES order number of the operation logged on to the workplace. If several operations are logged on to the workplace, you can select a different MES order number using a search help.

Input batch

In the field *Input batch*, enter the batch number of the input batch you want to log on in advance.

Buttons

Cancel

The button *Cancel* closes the dialogue without any further action.

Log on in advance

The selected input batch is logged on in advance.

Log off reserved batch

Function authorization	sma.cevab
------------------------	-----------

You use the posting dialog *Log off reserved batch* to log off an input batch from a current operation that has been logged on in advance/reserved.

If you call the function, an intermediate dialog opens. This intermediate dialog shows the component list of the current operation. Select an entry in the component list to log off the input batch logged on in advance/reserved.

Input fields

MES order number

The system enters the MES order number of the operation logged on to the workplace. If several operations are logged on to the workplace, you can select a different MES order number using a search help.

Input batch

In the field *Input batch*, enter the batch number of the batch you want to log off.

Buttons

Cancel

The button *Cancel* closes the dialogue without any further action.

Log off

The selected input batch, which has been logged on in advance, is logged off

Output batch change

Function authorization	sma.cawl
------------------------	----------

You use the posting dialog *Output batch change* to change the output batch for a current operation.

Display fields

Workplace

The system enters the selected workplace.

MES order number

The system enters the MES order number of the operation logged on to the workplace. If several operations are logged on to the workplace, you can select a different MES order number using a search help.

New batch

The field *New batch* displays the batch number of the next output batch.

Material

This field displays the material number of the output batch.

Input fields

MES order number

The system enters the MES order number of the operation logged on to the workplace. If several operations are logged on to the workplace, you can select a different MES order number using a search help.

Target buffer

This is the material buffer for the posting of the produced output batch.

Batch class

A list of possible batch statuses for the output batch is displayed. Select the batch status from the list.

Quantity

Input field to enter the produced quantity of the output batch.

Reason

In field *Reason*, you can enter a scrap reason with batch class = A. A list of possible scrap reasons is displayed. Select the scrap reason from the list.

Buttons

Cancel

The button *Cancel* closes the dialogue without any further action.

Change Output Batch

The output batch is changed.